

Table 1: Track-B `holdout_A` E0/E1/E2 operating points, fixed ϵ_S (population: `tight_exact4`, no mass-plane cut; `holdout_A` opened once, Phase-4M). R_B is seed mean $\pm$ std over 3 seeds; raw QCD is the seed-mean raw surviving-QCD count that sets the support tier ($\bullet \geq 100$, $\circ$ 10–99, $\times < 10$).

ϵ_S	Arm	AUC (secondary)	R_B (mean $\pm$ std)	raw QCD (mean)	support
0.60	E0	0.9729	105.65 ± 4.09	194.3	$\bullet$
0.60	E1	0.9769	130.19 ± 7.36	158.0	$\bullet$
0.60	E2	0.9739	88.88 ± 8.01	232.7	$\bullet$
0.50	E0	0.9729	173.74 ± 9.36	118.3	$\bullet$
0.50	E1	0.9769	237.57 ± 5.11	86.3	$\circ$
0.50	E2	0.9739	144.42 ± 12.26	143.0	$\bullet$
0.40	E0	0.9729	285.66 ± 43.88	73.3	$\circ$
0.40	E1	0.9769	439.52 ± 38.39	47.0	$\circ$
0.40	E2	0.9739	244.59 ± 19.30	84.3	$\circ$
0.25	E0	0.9729	736.37 ± 54.39	28.0	$\circ$
0.25	E1	0.9769	1193.18 ± 202.93	17.7	$\circ$
0.25	E2	0.9739	759.82 ± 124.63	27.7	$\circ$
0.20	E0	0.9729	1094.20 ± 133.80	19.0	$\circ$
0.20	E1	0.9769	1740.96 ± 240.23	12.0	$\circ$
0.20	E2	0.9739	1370.08 ± 501.83	17.0	$\circ$
0.10	E0	0.9729	4441.88 ± 483.21	4.7	$\times$
0.10	E1	0.9769	7972.61 ± 1610.71	2.7	$\times$
0.10	E2	0.9739	10250.50 ± 7248.20	3.0	$\times$

Table 2: Track-A model comparison at $\epsilon_S = 0.40$ (`post_training_evaluation_20260815_v2`, corrected/authoritative). Error bars on the two newly-trained models are seed-to-seed std (5-fold OOF $\times$ 3 seeds); the 68% bootstrap CI (primary seed, 2000 group-stratified replicates) is given separately. Frozen references have no interval computed in any artifact this table draws from.

Model	$R_B(\epsilon_S=0.40)$	68% bootstrap CI	status
Cut baseline ($R_{HH} < 34$)	1.95	–	frozen, single op. pt.
BDT CONTROL ₀ (34-feat.)	13.21	–	frozen ref. (recomp.)
Dense DNN	13.29 ± 0.26	[12.26, 14.22]	measured, this benchmark
CMS-inspired five-jet pairwise event Transformer	14.00 ± 0.30	[12.77, 14.92]	measured, this benchmark
FiveJetParTNet	14.15	–	frozen ref. (cited)
BDT NEW _C (55-feat.)	14.71	–	frozen ref. (recomp.)

Table 3: Track-A supported-rejection boundaries, fixed ϵ_B grid, primary seed (20260811) only. Support gate: `raw_background_rows` ≥ 100 AND `qcd_raw_rows` ≥ 10 AND `qcd_Neff` ≥ 10 (all three required). $\checkmark$ = gate passes; $\times$ = gate fails – not reported as a supported physical claim anywhere in this package.

Model	ϵ_B target	R_B	raw bkg. rows	qcd raw rows	qcd N_{eff}	gate
Dense DNN	0.1	10.0	95176	3134	580.90	$\checkmark$
Dense DNN	0.01	100.0	9518	345	41.69	$\checkmark$
Dense DNN	0.001	999.7	952	73	2.19	$\times$
Dense DNN	0.0001	10018.5	95	15	2.03	$\times$
CMS-inspired transformer	0.1	10.0	95176	4334	626.37	$\checkmark$
CMS-inspired transformer	0.01	100.0	9518	272	50.92	$\checkmark$
CMS-inspired transformer	0.001	999.7	952	23	6.56	$\times$
CMS-inspired transformer	0.0001	10018.5	95	2	1.13	$\times$

$\checkmark/\times$ require `pifont`. $\epsilon_B \in \{10^{-3}, 10^{-4}\}$ fail the gate for both models (`qcd_Neff` < 10) and must not be quoted as supported rejection numbers.

Table 4: Track-B `holdout_A` mechanism gain: seed-paired R_B ratios (mean $\pm$ std of the 3 per-seed ratios; support tier = the weaker of the two arms entering the ratio at that point).

ϵ_S	$R_B(E1)/R_B(E0)$			$R_B(E2)/R_B(E1)$		
	ratio	% change	supp.	ratio	% change	supp.
0.60	1.23 ± 0.09	+23.2%	●	0.69 ± 0.12	-31.7%	●
0.50	1.37 ± 0.09	+36.7%	○	0.61 ± 0.05	-39.2%	○
0.40	1.57 ± 0.31	+53.9%	○	0.56 ± 0.11	-44.4%	○
0.25	1.63 ± 0.35	+62.0%	○	0.65 ± 0.17	-36.3%	○
0.20	1.63 ± 0.44	+59.1%	○	0.84 ± 0.51	-21.3%	○
0.10	1.83 ± 0.60	+79.5%	×	1.17 ± 0.72	+28.6%	×

Ratio point estimate/std = mean/std of the 3 seed-paired ratios $R_B^{(i)}(\text{num})/R_B^{(i)}(\text{den})$; % change uses the ratio of seed means instead (matches the headline prose figures in `HOLDOUT_A_ADJUDICATION_RESULT.md` to within 3% relative outside the exploratory tier – see `VALIDATION_REPORT.json`).